

Software Requirements Specification

Restaurant Management Application

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Backend	Java Spring Boot
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Contents

1	Introduction	2
1.1	Purpose	2
1.2	Scope	2
1.3	Definitions	2
2	Overall Description	3
2.1	System Overview	3
2.2	User Roles	3
3	Functional Requirements	4
3.1	Authentication	4
3.2	User and Staff Management	4
3.3	Ingredient Management	4
3.4	Inventory Management	5
3.5	Recipe Management	6
3.6	Menu Item Management	6
3.7	Time Sheet and Clock In/Out	6
4	Data Model Summary	8

1 Introduction

1.1 Purpose

This document specifies the functional requirements for a web-based Restaurant Management Application. It serves as the reference for development, describing what the system must do from the perspective of its two user roles: Admin and Employee (Staff).

1.2 Scope

The application will allow a restaurant owner (Admin) to manage staff accounts, recipes, menu items, and inventory, while giving employees the ability to clock in/out and view relevant information. The system is a web application built with a Java Spring Boot backend and a relational database.

1.3 Definitions

Term	Definition
Admin	The restaurant owner or manager. Has full access to the system.
Employee	A staff member. Has limited, read/clock access.
Ingredient	A raw material used in recipes (e.g. eggs, chicken).
Inventory	Tracks the current stock quantity of each ingredient.
Recipe	A named set of ingredients with quantities and a preparation procedure.
Item	A menu item sold to customers. Linked to a Recipe and has a price.
TimeSheet	A record of an employee clock-in and clock-out for a given day.
Threshold	A minimum stock level for an ingredient. Triggers a low-stock alert when crossed.

2 Overall Description

2.1 System Overview

The Restaurant Management Application is a role-based web system. Upon login, users are directed to a dashboard appropriate to their role. The Admin dashboard provides full management capabilities. The Employee dashboard provides limited operational access.

2.2 User Roles

Role	Description
Admin	Full access: manage staff, recipes, inventory, items, and view reports.
Employee	Limited access: clock in/out, view assigned recipes, view inventory levels.

3 Functional Requirements

3.1 Authentication

ID	Requirement
FR-AU-01	The system shall allow any user to log in using a username and password.
FR-AU-02	The system shall reject login attempts with invalid credentials and display an error message.
FR-AU-03	The system shall assign a role (ADMIN or EMPLOYEE) to each user account.
FR-AU-04	The system shall restrict access to features based on the user role.
FR-AU-05	The system shall allow users to log out and terminate their session.

3.2 User and Staff Management

ID	Requirement
FR-US-01	The Admin shall be able to create new Employee accounts with a name, username, and password.
FR-US-02	The Admin shall be able to view a list of all Employee accounts.
FR-US-03	The Admin shall be able to deactivate or delete an Employee account.
FR-US-04	The Admin shall be able to reset an Employee password.

3.3 Ingredient Management

ID	Requirement
FR-IG-01	The Admin shall be able to create a new ingredient with a name, cost price, and unit of measurement.
FR-IG-02	The Admin shall be able to view a list of all ingredients.
FR-IG-03	The Admin shall be able to edit an ingredient name, cost price, or unit.
FR-IG-04	The Admin shall be able to delete an ingredient not currently used in any recipe.

ID	Requirement
FR-IG-05	Units of measurement shall be one of: PIECES, GRAMS, KILOGRAMS, LITERS, MILLILITERS, OUNCES.

3.4 Inventory Management

ID	Requirement
FR-IN-01	The system shall maintain an inventory record for each ingredient, tracking current quantity.
FR-IN-02	The Admin shall be able to update the quantity of any ingredient in inventory.
FR-IN-03	The Admin shall be able to set a low-stock threshold for each ingredient.
FR-IN-04	The system shall flag an ingredient as low-stock when its quantity falls at or below its threshold.
FR-IN-05	The Admin shall receive an in-app notification when an ingredient is flagged as low-stock.
FR-IN-06	The Employee shall be able to view the current inventory list (read-only).

3.5 Recipe Management

ID	Requirement
FR-RC-01	The Admin shall be able to create a new recipe with a name, preparation procedure, and a list of ingredients with required quantities.
FR-RC-02	The Admin shall be able to view all recipes.
FR-RC-03	The Admin shall be able to edit a recipe name, procedure, or ingredient quantities.
FR-RC-04	The Admin shall be able to delete a recipe not currently linked to a menu item.
FR-RC-05	Each ingredient in a recipe shall specify a quantity. The unit is inherited from the ingredient definition.
FR-RC-06	The Employee shall be able to view recipes (read-only).

3.6 Menu Item Management

ID	Requirement
FR-IT-01	The Admin shall be able to create a menu item with a name, price, and linked recipe.
FR-IT-02	The Admin shall be able to view all menu items.
FR-IT-03	The Admin shall be able to edit a menu item name, price, or linked recipe.
FR-IT-04	The Admin shall be able to delete a menu item.

3.7 Time Sheet and Clock In/Out

ID	Requirement
FR-TS-01	The Employee shall be able to clock in, recording the current date and time.
FR-TS-02	The Employee shall be able to clock out, recording the current time.
FR-TS-03	The system shall calculate and store the hours worked when the Employee clocks out (clock-out minus clock-in).
FR-TS-04	The Admin shall be able to view the timesheet records of all employees.

ID	Requirement
FR-TS-05	The Admin shall be able to view a biweekly hours summary per employee (sum of hours in a 2-week period).
FR-TS-06	The Employee shall be able to view their own timesheet history.

4 Data Model Summary

The following table summarizes the core entities and their key fields.

Entity	Key Fields
User	id, name, username, password (hashed), role (ADMIN / EMPLOYEE)
Ingredient	id, name, costPrice, unit (Enum)
Inventory	id, ingredient (FK), qty, threshold
Recipe	id, name, ingredients (Map: Ingredient → qty), procedure
Item	id, name, price, recipe (FK)
TimeSheet	id, user (FK), date, clockIn, clockOut, hours

Note: Fields marked FK are stored as foreign key IDs in the database. Spring Data JPA maps them to full Java objects at the application layer.